5 The Sylow Theorems

The Sylow Theorems provide a partial converse to Lagrange's Theorem. We start with a definition.

Definition 5.1.
Let p be a prime number. A finite group G of cardinality p^t for some $t \geq 1$ is called a (finite) *p-group*.

Let G be a finite group.

1. Subgroups of G which are p-groups are called *p-subgroups*.
2. If $|G| = p^r m$ with $\gcd(p, m) = 1$, then a p-subgroup of G has cardinality p^i with $1 \leq i \leq r$. A subgroup of G of cardinality p^r is called a *Sylow p-subgroup of G* or a *p-Sylow subgroup of G*.
3. The set of Sylow p-subgroups of G will be denoted $\mathrm{Syl}_p(G)$ and the cardinality of $\mathrm{Syl}_p(G)$, which is the number of Sylow p-subgroups of G, will be denoted

$$n_p(G) = |\operatorname{Syl}_p(G)|.$$

Example 5.1.

1. The group S_3 has $6 = 2 \times 3$ elements

$$S_3 = \{1, (123), (132), (12), (13), (23)\}.$$

The unique Sylow 3-subgroup of S_3 is

$$\langle (123) \rangle = \{1, (123), (132)\}$$

[this is the only subgroup of S_3 of cardinality 3], and the Sylow 2-subgroups of S_3 are

$$\{1, (12)\} \qquad \{1, (13)\} \qquad \text{and} \qquad \{1, (23)\}.$$

Thus $n_3(S_3) = 1$ and $n_2(S_3) = 3$.

2. The group D_{12} has $12 = 2^2 \times 3$ elements. It has three Sylow 2-subgroups, namely $\{1, r^3, s, sr^3\}$, $\{1, r^3, sr, sr^4\}$, and $\{1, r^3, sr^2, sr^5\}$, each of cardinality 2^2. It has many 2-subgroups: the Sylow 2-subgroups above, five subgroups $\langle sr^i \rangle$ for $i = 0, \ldots, 5$ and the subgroup $\langle r^3 \rangle$.
It has exactly one Sylow 3-subgroup, namely $\{1, r^2, r^4\}$. This is its only 3-subgroup.
None of the other subgroups of D_{12} are p-subgroups of D_{12}. For instance, $\langle r \rangle$, of cardinality 6, is not a p-subgroup of D_{12}, for any prime p.

Theorem 5.1 (The Sylow Theorems).
Let G be a finite group of cardinality $|G| = p^r m$ with $r \geq 1$ and $\gcd(p, m) = 1$.

(Sylow I) *Sylow p-subgroups of G exist, i.e. $\mathrm{Syl}_p(G) \neq \emptyset$.*

(Sylow II) *The Sylow p-subgroups of G are conjugate: if P_1 and P_2 are Sylow p-subgroups of G, then there exists $g \in G$ such that $P_1 = gP_2g^{-1}$.*

(Sylow III)

- *The number $n_p(G)$ of Sylow p-subgroups of G satisfies $n_p(G) \equiv 1 \mod (p)$.*
- *The number $n_p(G)$ of Sylow p-subgroups of G divides $m = \frac{|G|}{p^r}$.*
- *Moreover, $n_p(G) = |G : N_G(P)|$.*

We will extend Sylow I with Theorem 5.5, which states that every p-subgroup of G and every element of order a power of p is contained in a Sylow p-subgroup of G.

In order to prove the Sylow Theorems we add some complements to group actions in the case of an action of a p-group.

5.1 Action of finite p-groups

Let $(\cdot) : H \times X \to X$ be a group action where H is a p-group and X is a finite set. The Orbit-Stabiliser Theorem gives

$$|\operatorname{orb}(x)| = \frac{|H|}{|\operatorname{Stab}(x)|},$$

for any $x \in X$. In particular the following cases occur:

- either $|\operatorname{orb}(x)| > 1$ in which case $|\operatorname{orb}(x)|$ is divisible by p, since $|H|$ is a power of p and $|\operatorname{orb}(x)|$ divides $|H|$;
- or $|\operatorname{orb}(x)| = 1$. In this case $\operatorname{orb}(x) = \{x\}$ and we call x a *fixed point* of the given action of H.

We know

$$X = \bigcup_{1 \le i \le l} \operatorname{orb}(x_i)$$

is a finite union of disjoint orbits. Counting the number of elements of X as the sum of the cardinalities of the orbits $\operatorname{orb}(x_i)$ we have:

$$|X| = \sum_{i=1}^{l} |\operatorname{orb}(x_i)|.$$

Each $\operatorname{orb}(x_i)$ contributes to $|X|$ either by a power of p if x_i is not a fixed point of the action, or by 1 if x_i is a fixed point. In particular reducing modulo p, $|\operatorname{orb}(x_i)|$ reduces to 0 if x_i is not a fixed point, and we find that $|X|$ is congruent modulo p to the sum of 1's over the set of all fixed points of the action which is precisely the number of fixed points. This leads to the following important congruence modulo p for an action of a finite p-group.

Theorem 5.2 (Fixed Point Congruence)**.**
Let H be a finite p-group acting on a finite set X and $\operatorname{Fix}_H(X)$ the subset of fixed points of X under this action. Then

$$|X| \equiv |\operatorname{Fix}_H(X)| \mod (p).$$

Corollary 5.3.
Let H be a finite p-group acting on a finite set X. If $|X|$ is not divisible by p then there is at least one fixed point in X. If $|X|$ is divisible by p, then the number of fixed points is divisible by p (possibly equals 0).

Before embarking on the proof of Sylow I, we record the following useful lemma.

Lemma 5.4.
Let G be a group of cardinality $p^r m$. Let H be a subgroup of G of cardinality p^i, for some $1 \le i < r$. Then there exists a subgroup H' of G such that H is a subgroup of H' and $|H' : H| = p$ and hence $|H'| = p^{i+1}$.

Proof. Consider the left multiplication action

$$H \times G/H \to G/H$$

$$(h, gH) \mapsto hgH.$$

By the Fixed Point Congruence theorem, we have

$$|G/H| \equiv |\operatorname{Fix}_H(G/H)| \mod (p). \tag{*}$$

Now $gH \in \operatorname{Fix}_H(G/H)$ if and only if for all $h \in H$, we have $hgH = gH$. Equally, for all $h \in H$, $g^{-1}hg \in H$. Equally, $g^{-1}Hg \subset H$. Equally, $g^{-1}Hg = H$ (as $|g^{-1}Hg| = |H|$). Now,

$$g^{-1}Hg = H \iff gHg^{-1} = H \iff g \in N_G(H).$$

Thus,

$$\operatorname{Fix}_H(G/H) = \{gH : g \in N_G(H)\} = N_G(H)/H$$

and the above congruence (*) becomes

$$|G/H| \equiv |N_G(H)/H| \mod (p). \tag{**}$$

Now if $|H| = p^i$ with $1 \leq i < r$ then p divides $|G/H| = \frac{|G|}{|H|} = \frac{p^r m}{p^i} = p^{r-i}m$ hence divides $|N_G(H)/H|$ by (**). By Cauchy's Theorem $N_G(H)/H$ has a subgroup of order p which is of the form H'/H (as are all subgroups of $N_G(H)/H$) where H' is a subgroup of $N_G(H)$ containing H. Thus $|H' : H| = |H'/H| = \frac{|H'|}{|H|} = p$ which implies $|H'| = p|H| = pp^i = p^{i+1}$. □

Proof of (Sylow I). We show a stronger result namely that given any $1 \leq i \leq r$ there exists a subgroup of G of cardinality p^i. We prove this by induction on i. The case $i = 1$ is already established by Cauchy's Theorem.

Suppose that H is a subgroup of G of cardinality p^i. By Lemma 5.4, there exists a subgroup H' of G of cardinality p^{i+1}. The result follows by induction. □

Theorem 5.5 (Extension of Sylow I)**.**
Using Lemma 5.4, we may easily prove the following:

- *If H is a p-subgroup of G, then there exists a Sylow p-subgroup P of G such that $H \subset P$.*
- *An element of G whose order is a power of p is contained in a Sylow p-subgroup of G.*

Proof. For the first result, follow the proof of Sylow I above, but starting with H rather than a subgroup of cardinality p. For the second result, let x be an element of order p^i, for some $1 \leq i < r$. Then $x \in \langle x \rangle$. By the first part of this result, $\langle x \rangle$ belongs to some Sylow p-subgroup of G. □

Proof of (Sylow II). Let P and Q be Sylow subgroups of G, we show there exists $g \in G$ such that $Q = gPg^{-1}$.

Consider the action

$$Q \times G/P \to G/P$$

$$(g', gP) \mapsto g'gP.$$

Since Q is a finite p-group we have the Fixed Point Congruence

$$|G/P| \equiv |\operatorname{Fix}_Q(G/P)| \mod (p). \qquad (*)$$

As P is a Sylow p-subgroup of G, $|G/P| = m$ is not 0 modulo p, hence $|\operatorname{Fix}_Q(G/P)|$ is not congruent to 0 modulo p by (*). In particular, $|\operatorname{Fix}_Q(G/P)| \geq 1$ and $\operatorname{Fix}_Q(G/P) \neq \emptyset$.

Let $gP \in \operatorname{Fix}_Q(G/P)$, then $g'gP = gP$ for all $g' \in Q$. The latter implies $g'g \in gP$ for all $g' \in Q$, which implies $g' \in gPg^{-1}$ for all $g' \in Q$ hence $Q \subset gPg^{-1}$. This implies the equality $Q = gPg^{-1}$ since both Q and gPg^{-1} have the same cardinality p^r. □

Proof of (Sylow III). First, we prove $n_p(G) \equiv 1 \mod (p)$. Consider the action of a Sylow p-subgroup Q of G on $\operatorname{Syl}_p(G)$ by conjugation:

$$Q \times \operatorname{Syl}_p(G) \to \operatorname{Syl}_p(G)$$

$$(g, P) \mapsto gPg^{-1}.$$

The cardinality of $\operatorname{Syl}_p(G)$ is $n_p(G)$. Since Q is a finite p-group we have the Fixed Point Congruence

$$|\operatorname{Syl}_p(G)| \equiv |\operatorname{Fix}_Q(\operatorname{Syl}_p(G))| \mod (p). \qquad (*)$$

Fixed points of $\operatorname{Syl}_p(G)$ under the action of Q are Sylow p-subgroups P such that $gPg^{-1} = P$ for all $g \in Q$, one such P is Q itself. For all such P we have $Q \subset N_G(P)$ by the previous equality, and also $P \subset N_G(P)$. Thus both Q and P are Sylow p-subgroups of $N_G(P)$. Applying (Sylow II) to the group $N_G(P)$ we find that P and Q are conjugate in $N_G(P)$. Since P is normal in $N_G(P)$ the only subgroup of $N_G(P)$ that is conjugate to P is P itself and we obtain $Q = P$. Thus $\operatorname{Fix}_Q(\operatorname{Syl}_p(G)) = \{Q\}$ and $|\operatorname{Fix}_Q(\operatorname{Syl}_p(G))| = 1$, the congruence (*) above shows that $n_p(G) = |\operatorname{Syl}_p(G)| \equiv 1 \mod (p)$.

To show $n_p(G) = |G : N_G(P)|$, consider the action of G on $\operatorname{Syl}_p(G)$ by conjugation:

$$G \times \operatorname{Syl}_p(G) \to \operatorname{Syl}_p(G)$$

$$(g, P) \mapsto gPg^{-1}.$$

Let P be a chosen Sylow p-subgroup of G (this exists by Sylow I). Since all Sylow p-subgroups of G are conjugate by (Sylow II), this group action has only one orbit which is necessarily equal to $\operatorname{Syl}_p(G) = \operatorname{orb}(P)$. By the Orbit–Stabiliser Theorem,

$$n_p(G) = |\operatorname{Syl}_p(G)| = |\operatorname{orb}(P)| = \frac{|G|}{|\operatorname{Stab}_G(\{P\})|} = \frac{|G|}{|N_G(P)|} = |G : N_G(P)|.$$

Now P is a subgroup of $N_G(P)$, so

$$p^r m = |G| = |G : N_G(P)|\,|N_G(P)| = n_p(G)|N_G(P) : P|\,|P| = n_p(G)\,|N_G(P) : P|\,p^r,$$

whence $m = |N_G(P) : P| n_p(G)$, so $n_p(G)$ divides m. □

Corollary 5.6.
Let P be a Sylow p-subgroup of a finite group G. Then P is the unique Sylow p-subgroup of G, i.e. $n_p(G) = 1$, if and only if P is a normal subgroup of G.

Proof. Note that every conjugate gPg^{-1} of P is a Sylow p-subgroup, and the converse holds by Sylow II. Thus

$$\mathrm{Syl}_p(G) = \{gPg^{-1} : g \in G\}.$$

Now $n_p(G) = 1$, i.e. $\mathrm{Syl}_p(G) = \{P\}$ if and only if $gPg^{-1} = P$ for every $g \in G$, which is equivalent to P being a normal subgroup. □

Example 5.2.
(1) A finite abelian group A has a unique Sylow p-subgroup $A_p \subset A$ for each prime number p dividing its cardinality since all subgroups of A are normal. The subgroup A_p consists of all elements $x \in A$ whose order is a power of p.

If $G = \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/q^s\mathbb{Z}$ with p and q distinct primes then $\mathbb{Z}/p^r\mathbb{Z} \times \{0\}$ is the Sylow p-subgroup of G and $\{0\} \times \mathbb{Z}/q^s\mathbb{Z}$ is the Sylow q-subgroup of G.

(2) $|A_4| = 12 = 3 \times 2^2$. The alternating group A_4 has a unique Sylow 2-subgroup: $V_4 = \{1, (12)(34), (13)(24), (14)(23)\}$, it is a normal subgroup of A_4. A Sylow 3-subgroup of A_4 has cardinality 3 and is generated by a 3-cycle, there are four Sylow 3-subgroups: $\langle(123)\rangle$, $\langle(134)\rangle$, $\langle(124)\rangle$, $\langle(234)\rangle$ in A_4. Note that $4 \equiv 1 \mod (3)$.

(3) $|S_4| = 24 = 3 \times 2^3$. Prove that $n_2(S_4) = 3$, $n_3(S_4) = 4$, and find all the Sylow 3-and Sylow 2-subgroups of S_4.

The Sylow Theorems are useful in classifying groups of small cardinality and groups whose cardinality has few prime factors.

Groups of order pq, where p and q are distinct primes

Suppose $|G| = pq$ for primes p and q with $p < q$. Let $P \in \mathrm{Syl}_p(G)$ and $Q \in \mathrm{Syl}_q(G)$. We show that Q is normal in G; i.e. $n_q(G) = 1$, and P is normal in G if p doesn't divide $q - 1$. We have $n_q(G) = 1 + kq$ for some $k \geq 0$, $n_q(G)$ divides p and $p < q$. This forces $k = 0$. Since $n_q(G) = 1$, Q is the unique Sylow q-subgroup of G and is a normal subgroup of G. Since $n_p(G)$ divides q we have $n_p(G) = 1$ or q. Further $n_p(G) \equiv 1 \mod (p)$. So if q is not congruent to 1 modulo p; meaning p doesn't divide $q - 1$, then $n_p(G) = 1$ necessarily and P is a normal subgroup of G.

For example if $p = 5$ and $q = 13$ then a finite group with cardinality $65 = 5 \times 13$ has a unique Sylow 5-subgroup and a unique Sylow 13-subgroup since 13 is not congruent to 1 modulo 5.

Suppose $p = 3$ and $q = 7$. A finite group G of cardinality $21 = 3 \times 7$ has a unique Sylow 7-subgroup H which is normal in G. Further $n_3(G)$ is congruent to 1 modulo 3 and divides 7 hence equals either 1 or 7.

Groups of order 12

Let G be a finite group of order $12 = 3 \times 2^2$. We show that either G has a normal Sylow 3-subgroup or G is isomorphic to A_4 (in the latter case G has a normal Sylow 2-subgroup). Suppose $n_3(G) \neq 1$ and let $P \in \mathrm{Syl}_3(G)$. Since $n_3(G)$ divides 4, and $n_3(G)$ is congruent to 1 modulo 3, we must have $n_3(G) = 4$. As $|G| = 3 \times 2^2$, the Sylow 3-subgroups have cardinality 3, so the intersection of distinct Sylow 3-subgroups is the trivial group. Each Sylow 3-subgroup contains two elements of order 3. So G contains 2×4 elements of order 3. Since $|G : N_G(P)| = n_3(G) = 4$, by (Sylow III), we must have $N_G(P) = P$ as $P \subset N_G(P)$ and they have the same cardinality: $|N_G(P)| = \frac{|G|}{|G:N_G(P)|}$.

Now G acts on its 4 Sylow 3-subgroups $\{P_1, P_2, P_3, P_4\}$ by conjugation, so this action affords a representation

$$\phi : G \to S_4.$$

The kernel K of this action is the subgroup of G which normalises all Sylow 3-subgroups of G, i.e.

$$K = \bigcap_{i=1}^{4} N_G(P_i) = \bigcap_{i=1}^{4} P_i.$$

In particular $K = \{1\}$, since $P_i \cap P_j = \{1\}$ if $i \neq j$, i.e. ϕ is injective and G is isomorphic to its image $\phi(G) \subset S_4$. Since G has 8 elements of order 3 and there are precisely 8 elements of order 3 in S_4, all contained in A_4, it follows that $\phi(G)$ intersects A_4 in a subgroup of order at least 8. Since both groups have order 12 it follows that $\phi(G) = A_4$, so that G is isomorphic to A_4.

Groups of order p^2q, where p and q are distinct primes

Suppose $|G| = p^2q$ for distinct primes p and q. We show G has a normal Sylow subgroup for either p or q. Let $P \in \mathrm{Syl}_p(G)$ and $Q \in \mathrm{Syl}_q(G)$. Consider the case $p > q$. Since $n_p(G)$ divides q and $n_p(G) = 1 + kp$ for some $k \geq 0$, we must have $n_p(G) = 1$ and P is a normal subgroup of G in this case. Consider now the case $p < q$. If $n_q(G) = 1$, Q is normal in G. Assume therefore that $n_q(G) > 1$, i.e. $n_q(G) = 1 + tq$ for some $t > 0$. Now $n_q(G)$ divides p^2 so $n_q(G)$ equals p or p^2. Since $q > p$ we can not have $n_q(G) = p$ so $n_q(G) = 1 + tq = p^2$. Thus

$$tq = p^2 - 1 = (p-1)(p+1).$$

Since q is prime, either q divides $p - 1$ or q divides $p + 1$. The former is impossible since $q > p$ so the latter holds. Since $q > p$ and q divides $p + 1$ we must have $q = p + 1$ which only holds if $p = 2$, $q = 3$, and $|G| = 12$. The result now follows from the previous example which shows G is isomorphic to A_4 in this case and the Sylow 2-subgroup of G is normal in G.